Operational challenges also emerge. In dynamic systems, entities and relationships are often created and removed, making lifecycle management essential. Because access logic is embedded in the graph's structure, even minor changes can have unintended consequences, especially in complex systems where relationship pathways aren’t obvious. While ReBAC allows teams to author policies through user-friendly interfaces, that same flexibility can complicate governance and auditing without specialized tools. Organizations adopting ReBAC should invest in guardrails, training, and simulation tools to keep the system safe and predictable.

ReBAC works well when access is based on how people naturally collaborate. However, when access depends on what a person is doing—not just who they’re connected to—other models, such as ABAC, might be a better choice.

6.2 Attribute-based access control

As ACME Corp grew, its internal systems and the teams using them needed increasingly fine-grained, context-aware access controls. Engineering required secure deployment pipelines that allowed changes only under specific conditions. Finance systems stored confidential reports that needed clearance-based restrictions. HR systems had to expose sensitive records to managers, but only in certain situations. Relying solely on roles or relationships was not sufficient. Access often depended on the attributes of the person making the request, the resource being accessed, and the conditions under which the request occurred.

Attribute-Based Access Control (ABAC) provided ACME with the flexibility it needed. ABAC makes decisions based on attributes, including named values linked to the principal (such as department or clearance), the resource (like classification or owner), the environment (for example, time or network risk), or the request itself (such as action type). Instead of tying access directly to roles or relationships, ABAC employs policies that are evaluated using these attributes at the time of the decision.

This model proved useful across the organization: